Zhalok Rahman

Software Engineer 2 at Datatree Technology
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LinkedIn: https://www.linkedin.com/in/zhalok-rahman/
Github: https://github.com/zhalok
Video Introduction: https://bit.ly/4jpZSJo

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline

Education

Shahjalal University of Science and Technology
Bachelors of Science in Computer Science
2019-01-01 – 2024-04-01

Experience

Kinetik Healthcare *Software Engineer 2*
2025-04-01 – Present

- Built realtime features with **Redis Pub/Sub** and **Websocket**
- Migrated legacy **ExpressJS** service to **NestJS**
- Architected async event driven solutions using **CDC Streams**, **Amazon SQS**, **Kafka**, **Redis streams**
- Managed **PostgreSQL** database in **Amazon RDS**
- Improved **MongoDB** performance at scale
- Containerized and deployed applications to **Amazon ECS**
- Used **Amazon ECR** for managing container images
- Provisioned AWS infrastructure as code using **CloudFormation**
- Deployed static resources to **Amaazon Cloudfront**
- Configured DNS in **Amazon Route 53**
- Automated deployment with **Amazon Codepipeline**
- Built **Grafana** dashboards for visualizing and aggregating metrics, logs and traces from multiple sources
- Set alerts for detecting downtimes and failures upfront
- Investigated incidents and fixed root causes based on **Logs traces and application event**
- Built tools for internal testing and monitoring
- Conducted load testing to ensure system reliability at scale before releasing to production
- Writen unit tests to ensure code reliability
- Automated build check, unit test check in continious integration pipeline
- Conducted spec driven development for AI assisted coding
- Managed SDLC with isolated dev, stage and production environments

PropertyFinder *Software Engineer*
2024-10-01 – 2025-10-31

- Ensured reliability and performance of **Golang** based B2B platform
- **Traced race conditions** between distributed services that was causing p1 incidents and confusing user experiences
- Investigated and fixed root cause of critical production incidents using metrics, logs, and traces from **Grafana**
- Conducted **Load Testing** and built dashboards to observe the performance of the entire system under high stress
- Written scripts for data fix to mitigate incidents
- Contributed to the modernization of B2B service, making architectural decisions and strategies for efficient migration
- Written unit tests to ensure **80 percent coverage**
- Designed asynchronous event-driven pipelines using **SQS** and **Kinesis Streams**
- Configured alerts in **Grafana** with **Slack** integration for real-time notification on any potential incident
- Used **Windsurf** for studying legacy codebases efficiently

Learnyx AI *Fullstack Engineer*

2023-10-01 – 2025-02-01

- Architected and built a scalable REST API with **NestJS**
- Designed an AI generation pipeline using **LangChain** for structured prompt management and multi-step chaining
- Integrated **Langfuse** for LLM observability, enabling cost tracking and performance tracing across model calls
- Decoupled heavy workloads from the main request cycle using asynchronous **RabbitMQ** consumers, improving resource efficiency
- Enhanced UX by implementing real-time streaming of AI-generated responses
- Optimized load performance using a hybrid lazy-loading and preloading strategy
- Acted as a technical advisor in stakeholder discussions, bridging engineering constraints with business goals
- Mentored a team of 3 engineers on system architecture patterns and backend best practices
- Managed self-hosted infrastructure on **AWS EC2** using **Coolify**, supporting **500+ active users**
- Integrated **Stripe** to handle recurring billing, subscription lifecycle, and payment management

Projects

MongoDB vs PostgreSQL benchmarking for storing documents
Benchmark harness comparing MongoDB BSON and PostgreSQL JSONB for document storage under concurrent load, with container and database-internal metrics.

- Benchmark results: https://bit.ly/4dbwS4h
- Github: https://bit.ly/3TYSpGZ

Tech: Golang, JavaScript, Docker, PostgreSQL, MongoDB, Prometheus, Grafana, Loki, Promtail

PostgreSPECTOR
Observability toolkit for PostgreSQL that exposes execution stats and container metrics to Prometheus and Grafana, with logs shipped via Loki.

- Demo: https://youtu.be/w2OCn0X3bRw
- Github: https://bit.ly/4y8AFXX

Tech: Golang, Prometheus, Grafana, PostgreSQL, Loki, Promtail, Docker

Obstainer
Lightweight container observability tool that reads metrics straight from cgroups and exposes them to Prometheus and Grafana.

- Demo: https://youtu.be/sifxPdFs2AE
- Github: https://bit.ly/4y8AL1L

Tech: Golang, Grafana, Prometheus, Docker, Loki, Promtail

Data Stores Python
Experiments comparing TCP server I/O models (thread-per-connection, non-blocking, event-driven) under load.

- Benchmark results: https://bit.ly/3V6UUr9
- Github: https://bit.ly/4xMsr7q

Tech: Python, Socket, System Calls, Docker

Go Http From Scratch System Calls
HTTP server implemented from raw sockets and system calls in Go, without the net/http package.

- Github: https://bit.ly/4z6JOBb

Tech: Golang, Syscalls, OSSignals, Sockets

Certifications

AWS Certified Cloud Practitioner
AWS
https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e
Deep dive on AWS services

Fundamentals of Operating Systems
Udemy
https://www.udemy.com/certificate/UC-ff0d44a3-8917-40cd-aef9-a01e258aee49/
Learned about Process structure, Memory managment, CPU cycles, I/O management

Fundamentals of Network Engineering
Udemy
https://www.udemy.com/certificate/UC-0a49d23e-036a-4c80-ad5a-410ccb877ebb/
Learned about under the hood mechanisms of network protocols

Fundamentals of Database Engineering
Udemy
https://www.udemy.com/certificate/UC-afe8cbda-1420-4cca-a833-aed013e7a6e2/
Learned about ACID, Data organization, Indexing, Partitioning, Sharding, Concurrency control, Replication, Sharding